

This metric can also be formulated for buy and sell orders as follows:

$$RPM_{Buy} = \frac{1}{2} \cdot \left(\frac{Total Volume + Volume at Price > P_{avg} - Volume at Price < P_{avg}}{Total Volume} \right) \quad (3.18)$$

$$RPM_{Sell} = \frac{1}{2} \cdot \left(\frac{Total Volume + Volume at Price < P_{avg} - Volume at Price > P_{avg}}{Total Volume} \right) \quad (3.19)$$

This formulation of RPM is now the average of the percentage of volume that traded at our execution price or better and 1 minus the average of the percentage of volume that traded at our price or worse. Thus, in effect, it treats half of the investor's orders as better performance and half the order as worse performance. As stated, the original formulation treated all of the investor's shares as better performance and inflated the measure.

The RPM is in many effects a preferred measure to the VWAP metric because it can be used to compare performance across stocks, days, and volatility conditions. And it is not influenced to the same extent as VWAP when large blocks trade at extreme prices.

The RPM will converge to 50% as the investor accounts for all market volume in the stock on the day similar to how the VWAP converges to the average execution price for large orders.

Brokers achieving fair and reasonable prices on behalf of their investors should achieve an RPM score around 50%. RPM scores consistently greater than 50% are an indication of superior performance and scores consistently less than 50% are an indication of inferior performance. The RPM measure can also be mapped to a qualitative score, for example:

RPM	Quality
0–20%	Poor
20–40%	Fair
40–60%	Average
60–80%	Good
80–100%	Excellent

Pre-Trade Benchmark

The pre-trade benchmark is used to evaluate trading performance from the perspective of what was expected to have occurred. Investors compute the difference between actual and estimated to determine whether performance was reasonable based on how close they came to the expectation.